MILITARY SPECIFICATION

PROJECTILE, BODY, SMOKE, WP, 81MM, M370 METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Projectile, Body, Smoke, WP, 81mm, M370, Metal Parts Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 —Boxes, Fiber Board

MILITARY

MIL-B-121 —Barrier Material, Greasproofed, Waterproofed, Flexible

MIL-A-2550 —Ammunition and Special Weapons, General Specifications for

MIL-P-10025—Packaging, Packing and Marking for Interplant Shipment of Inert Ammunition Components: General Specification for

MIL-I-45208 —Inspection System Requirements

STANDARDS

FEDERAL

Federal Test Method Standard No. 151 —Metals; Test Methods

MILITARY

MIL-STD-105 —Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-109 —Quality Assurance Terms and Definitions

MIL-STD-639 —Surface Roughness Values Preferred for Army Ordnance Materiel

MIL-STD-1235 —Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

MUNITIONS COMMAND

C10520016 —Metal Parts Assembly
F7548604 —Pallet
A8651379 —Miniature Inspection Lamp

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(Copies of specifications, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Materials, parts and assemblies shall comply with the requirements specified on the applicable drawing, referenced specifications and the following:

3.2 Initial Production sample. Unless otherwise directed by the contracting officer, an initial production sample shall be required.

3.3 Heat treatment. Where applicable, each projectile body in a lot shall be heated above the upper critical temperature point, oil quenched and tempered uniformity to meet the mechanical properties specified on the applicable drawing. No work involving plastic flow is permitted after tempering unless followed by a stress relief.

3.4 Stress relief. Cold worked bodies shall be stress relieved at a minimum temperature of 700° Fahrenheit (F.) . No work involving plastic flow of metal is permitted after final stress relief.

3.5 Mechanical properties. All components shall meet the mechanical properties requirements as specified on the applicable drawings.

3.6 Metal defects. All components shall be free from cracks, seams, splits, cold shuts, inclusions, porosity or any similar defect.

3.7 Brazed joint. Prior to painting of the projectile assembly, the braze metal shall be visible for the full 360° around the outside of the joint between the adapter and the forward end of the body.

3.7.1 *Joint section.* When sectionalized, the braze joint shall be free of voids, cracks, entrapped slag or other foreign matter. The sum total of these defects shall not exceed 15 percent of the total exposed brazed joint.

3.8 Functional.

3.8.1 *Security.* There shall be no breakup or deformation of the projectile body assembly in the mortar tube or in flight.

3.9 Workmanship. The requirements for workmanship are as shown on the applicable drawings in the applicable specifications and the following:

3.9.1 *Surface finish.* The requirements for surface finish are as detailed on the applicable drawing. The cavity shall be free of burrs, scale, fins, deep seams, pits or other surface discontinuities.

3.9.2 *Protective coating.* Protective coating shall be in accordance with the applicable drawing and the specification referenced thereon.

3.9.3 *Burr.* No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.9.4 *Foreign matter.* No part or assembly shall contain dirt, grease, chips, rust, corrosion, or other foreign matter. Particular attention shall be paid to the cavity of the body assembly, prior to painting, to insure that no grit, scale or other foreign matter remains therein after final cleaning.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purhase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Quality assurance terms and definitions.* Reference shall be made to Standard MIL–STD–109 to define quality assurance terms used.

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4.1.2 *Inspection.* Inspection shall be in accordance with Specifications MIL–A–2550 and MIL–I–45208, except as specified herein.

4.2 **Initial production sample.** Prior to the initiation of regular production, the contractor shall submit an initial production sample to a Government-approved facility for evaluation. The sample shall consist of 70 body assemblies as follows: 55 body assemblies complete, including paint, 5 body assemblies unpainted, 10 sets of component parts unassembled. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, must be the same as will be used for regular production and shall be obtained from the same source of supply. The contractor shall inspect the sample in accordance with his inspection system and other inspection requirements of Specification MIL–I–45208. After inspection and provisional acceptance at source, the initial production sample shall be inspected at the Government-approved facility for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted. Fifty of the above assemblies shall be forwarded by the contractor to a designated proving ground for ballistic testing.

4.3 **Inspection provisions.**

4.3.1 *Lot.*

4.3.1.1 *Submission of product.* The product shall be submitted in accordance with Standard MIL–STD–105 or Standard MIL–STD–1235, as applicable.

4.3.1.2 *Heat treated or stress relieved lots.* Heat treated or stress relieved lots shall each be from one heat of steel.

4.3.1.3 *Cold shaping lot.* Cold shaping lots shall contain bodies from not more than one heat of steel, except in those instances where the manufacturer, with prior approval of the procuring activity, has demonstrated that his process is capable of producing acceptable components from steel within a specified chemistry range.

4.3.1.4 *Miscellaneous lot.* Small lots, such as bodies from abnormally small heats of steel or bodies which have become separated from previously delivered lots, etc., may, at the discretion of the manufacturer, be grouped into miscellaneous, forging or cold shaping lots as the case may be. Such lots shall be identified by a miscellaneous lot code number.

4.3.1.5 *Grouping.* Heat treated or stress relieved lots of forging or cold shaping lots shall remain grouped together for lot identity purposes, according to forging or stress relieved lots, until such time as the results of the tensile test have indicated that the lot has complied with the mechanical properties requirement of the applicable drawing.

4.3.1.6 *Projectile lot group.* Bodies of heat treated or stress relieved lots that have been determined to have complied with the mechanical properties requirements of the applicable component drawing, may be regrouped for assembly into projectile lot groups. A projectile lot group shall consist of projectiles produced by one manufacturer, under one contract, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision.

4.3.2 *Examination.* One hundred per cent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL–STD–105 or Standard MIL–STD–1235, as applicable. An AQL of 0.25 per cent for each major defect and an AQL of 0.40 per cent for each minor defect shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter. The Ordnance Standard Finish Blocks prescribed by Standard MIL–STD–639 shall be used as a basis of comparison for surface roughness determination. The miniature Inspection Lamp, Dwg.

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A8651379 or equal, shall be used for examination to determine tangency of radius at mouth of small inside diameter.

4.3.3 *Classification of defects.*

4.3.3.1 *Adapter, Dwg. F10520024, covering a detail of Dwg. C10520016.*

	Defect	Method of inspection	
CRITICAL:			
1.	Scoring or longitudinal scratches on wall of small inside diameter	Visual	
2.	Metal defective	Visual	
MAJOR:			
101.	Small inside diameter not functionally concentric with pitch diameter of thread	Gage	
102.	Small inside diameter	Gage	
103.	Diameter of counterbore at small inside diameter	Gage	
104.	Depth from front face to bottom of fuze cavity	Gage	
105.	Length to datum diameter	Gage	
106.	Tangency of radius at mouth of small inside diameter	Visual	(See 4.3.2)
107.	Surface finish on wall of small inside diameter	Visual & measure	
MINOR:			
201.	Diameter at front	Gage	
202.	Diameter at shoulder	Gage	
203.	Diameter of brazing area	Gage	
204.	Diameter at rear of brazing area	Gage	
205.	Depth from front face to start of angle in fuze cavity	Gage	
206.	Length total	Gage	
207.	Length from front face to shoulder	Gage	
208.	Length of brazing area	Gage	
209.	Presence of thread chamfer	Visual	(See 6.2)
210.	Radii or chamfer, missing or incorrect	Visual	(See 6.2)
211.	Surface finish improper, unless otherwise classified	Visual	(See 6.2)
212.	Burr	Visual	
213.	Foreign matter	Visual	

4.3.3.2 *Body, Dwg. F10520020, covering a detail of Dwg. 10520016.*

	Defect	Method of inspection	
CRITICAL:			
	None defined		
MAJOR:			
101.	Concentricity of cavity with pitch diameter of thread	Gage	
102.	Diameter of bourrelet, min.	Gage	
103.	Diameter of cavity	Gage	
104.	Length to forward datum diameter	Gage	
105.	Length to center datum diameter	Gage	
106.	Length to rear datum diameter	Gage	
107.	Wall thickness at ogive (min.)	Gage	
108.	Wall thickness at rear datum diameter	Gage	
109.	Wall thickness at datum length	Gage	
110.	Wall thickness variation (max.)	Gage	
111.	Thickness at juncture of thread boss and shoulder (min.)	Gage	
112.	Thickness through base	Gage	
113.	Surface finish, improper	Visual	(See 6.2)
114.	Metal defective	Visual	
115.	Diameter Brazing Surface	Gage	
MINOR:			
201.	Diameter at front exterior	Gage	
202.	Diameter at rear of body	Gage	

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	Defect	Method of inspection
203.	Length from front face to rear shoulder	Gage
204.	Length total, max.	Gage
205.	Length from rear to shoulder	Gage
206.	Radii or chamfers missing or incorrect	Visual (See 6.2)
207.	Burr	Visual
208.	Foreign matter	Visual

4.3.3.3 *Body, Assembly, Dwg. F10520019, prior to paint, covering a detail of Dwg. C10520016-*

	Defect	Method of inspection
CRITICAL:		
	None defined.	
MAJOR:		
101.	Concentricity of ogive with pitch diameter of rear thread, including perpendicularity of rear shoulder	Gage
102.	Concentricity of bourrelet with pitch diameter of rear thread, including perpendicularity of rear shoulder	Gage
103.	Concentricity of bourrelet with pitch diameter of nose thread including perpendicularity of front face	Gage
104.	Concentricity at rear with pitch diameter of rear thread, including perpendicularity of rear shoulder	Gage
105.	Length total, max.	Gage
106.	Air pressure test stamp missing (indicating failure to comply with requirements)	Visual
107.	Braze metal not visible for full 360° around joint between body and adapter	Visual
108.	Body protruded beyond adapter	Visual & Measure
109.	Foreign matter in cavity	Visual
MINOR:		
201.	Marking missing, incorrect or illegible	Visual

4.3.3.4 *Body, Assembly, Dwg. F10520019, after paint, covering a detail of Dwg. C10520016.*

	Defect	Method of inspection
CRITICAL:		
	None defined.	
MAJOR:		
101.	Weight	Approved scale
102.	Pitch diameter of nose thread, max.	Gage
103.	Minor diameter of nose thread, max.	Gage
104.	Pitch diameter of rear thread, min.	Gage
105.	Major diameter of rear thread, min.	Gage
106.	Diameter of bourrelet, max.	Gage
107.	Length of effective thread, min.	Gage
108.	Protective coating in cavity inadequate or excessive	Visual
109.	Pool of paint in cavity	Visual
MINOR:		
201.	Pitch diameter of nose thread, min.	Gage
202.	Pitch diameter of rear thread, max.	Gage
203.	Length total, max.	Gage
204.	Protective coating on inside of adapter or on base thread	Visual
205.	Paint not dry	Visual & Tactual
206.	Exterior paint, inadequate or damaged	Visual
207.	Silicone compound missing from indicated surfaces	Visual
208.	Burr	Visual
209.	Foreign matter	Visual

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4.3.3.5 *Metal Parts Assembly, Dwg. C10520016.*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
101. Length from front of cap to shoulder (max.)	Gage
102. Protector cap, missing or damaged	Visual
MINOR:	
None defined.	

4.3.3.6 *Pallet, Dwg. F7548604.*

Defect	Method of inspection
CRITICAL:	
None defined.	
MAJOR:	
None defined.	
MINOR:	
201. Strapping missing, loose or broken	Visual & Manual
202. Pallet damaged	Visual
203. Improperly packed	Visual
204. Marking missing, incorrect or illegible	Visual

4.3.4 *Inspection testing.* The following tests shall be performed in accordance with the provisions of 4.4, except as provided in Specification MIL–I–45208.

4.3.4.1 *Mechanical properties.*

4.3.4.1.1 *Yield strength and elongation test.* After final heat treatment or stress relief, two bodies from each heat-treated or stress-relieved lot shall be subjected to this test to determine compliance with the yield point and elongation requirements of the applicable drawing. Failure of any test specimen to comply shall be cause for rejection of the lot.

4.3.4.1.2 *Air pressure test (A major defect).* Each body assembly shall be subjected to a submerged air pressure test to determine compliance with the assembly drawing. Failure of any assembly to comply as evidenced by the presence of air bubbles in the vicinity of the brazed joint or rising to the surface of the water shall be cause for rejection of the assembly. Body assemblies which comply shall immediately upon removal from the test fixture be stamped with the letter "A" as indicated on the assembly drawing.

4.3.4.1.3 *Sectionalized joint.* After air pressure testing, but prior to the application of protective coating, 2 body assemblies from each lot shall be selected at random and sectionalized to determine compliance with 3.7.1. Failure of any specimen to comply shall be cause for rejection of the lot.

4.3.4.2 *Protective coating.* The sample size and acceptance criteria shall be in accordance with the assembly drawing and the requirements of the specifications listed thereon. Failure to comply with the requirements of the applicable specifications shall be cause for taking the corrective action recommended therein.

4.3.5 *Functional.* The following proving ground tests will be performed by the Government:

4.3.5.1 *Initial production sample.* The quantities listed in table I shall be ballistically tested in the phases indicated.

TABLE I

Phase	Quantity
Recovery	25
Velocity-Accuracy	25

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4.3.5.2 *Production lot sample.* The quantities listed in table II shall be ballistically tested in the phases indicated.

TABLE II

Phase	*Quantity*
Recovery	10 each lot.
Velocity-Accuracy	10 each lot (see 6.1)

Failure of any round fired for recovery to comply with 3.8.1 shall be cause for rejection of the lot. Velocity-Accuracy data will be recorded for informational purposes only.

4.3.6 *Inspection equipment.* The examination and tests shall be performed in accordance with the requirements of MIL–I–45208.

4.4 Test Methods and procedures.

4.4.1 *Mechanical properties.*

4.4.1.1 *Yield strength and elongation.* From the bodies selected in accordance with 4.3.4.1.1, two tension test specimens, conforming as closely as possible to the largest type "R" specimen obtainable as prescribed by Federal Test Method Standard No. 151, shall be taken longitudinally from each side of the body 180° apart. Procedures for determining yield strength and elongation shall be as prescribed by Federal Test Method Standard No. 151.

4.4.2 *Air pressure test.* This test shall be made using equipment capable of performing the test accurately and consistently. Body assemblies shall be mounted in an approved fixture in such a manner that the braze joint shall be submerged in water, the required air pressure shall be applied and held for a minimum of 30 seconds. The water in the tank shall be permitted to settle after immersion of the assemblies prior to the start of timing. The equipment shall be calibrated at the start of each day's operation and at intervals of 4 hours and after a stoppage of 3 hours or more.

4.4.3 *Sectionalized joint.* The samples, selected in accordance with 4.3.4.1.3, shall be sectionalized as follows: (a) The body portion of the projectile assembly to the rear of the nose adapter braze shall be removed in such a manner that no portion of the braze joint will be damaged; (b) the remaining portion of the body containing the nose adapter shall then be sectioned into four segments by cutting vertically through the adapter and body; (c) the brazed areas of the four pieces of brazed adapter and body shall be visually examined to determine compliance with 3.7.1.

4.4.4 *Protective coating.* The procedures for testing the cleaning of components and application of protective coating shall be in accordance with the requirements of the assembly drawing and the criteria for testing as specified in the applicable specifications.

4.4.5 *Functional.*

4.4.5.1 *Recovery.* The samples furnished in accordance with 4.3.5.1 or 4.3.5.2 inert loaded to the service weight, will be assembled into complete rounds utilizing the M141 fin assembly and fired in a mortar for which the round in standard. The propellant charge will be adjusted to obtain 112 percent of the maximum rated pressure of the round in a new mortar.

4.4.5.2 *Velocity-accuracy.* The samples, furnished in accordance with 4.3.5.1 or 4.3.5.2 inert loaded, will be assembled into complete rounds utilizing the service charge and fired into a mortar for which the round is standard for comparison with an equal number of rounds from a reference lot. The sample round and reference rounds will be fired alternately. Horizontal and vertical distances will be recorded for each round fired for determination of horizontal and vertical probable error (PE) in mils to assess accuracy. Data obtained will be recorded for information only and forwarded to the cognizant technical agency.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 *Level A.* Not applicable

5.1.2 *Level B.* Not applicable

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5.1.3 *Level C.* Bodies shall be packed in accordance with the requirements of Specification MIL–P–10025.

5.2 **Pallet.** Shall be in accordance with Dwg. F7548604, Sheets I and II. For dimensions of pallet, see table below:

Unit Weight	A	B	C	D	E	F	G	H
6.36 lbs.	36⅛	40⅜	36⅛	40⅜	10⅝	5⅝	33	40⅜

No. Per Pallet 100

Approx. Wt. of packed pallet 750 lbs.

2. Half-Slotted Partition—Type II, Class 1, solid fiberboard, 500 lbs. min. test board specification PPP–B–636. Construct in a 10 x 10 pattern, 100 compartments, each 3.2 square. Assembled dimensions 32⅞ x 32⅞ x 10 5/16. One required per pallet.

3. Filler—Same material as half-slotted partition 33. X 33. One or more required per pallet.

4. Liner—Barrier material, Grade A. Class 1, Type II, Specification MIL–B–121. 33. X 33. One required per pallet.

B. Packing Instructions

Pack 100 rounds in accordance with packing instructions on drawing F7548604, Sheet 1, except:

Place a filler on base of pallet and a liner on top of filler. Insert half-slotted partition and pack rounds with shipping cap upward. Add fillers along top, sides and end for tight packing (if possible, place one filler along each face of pallet).

C. Marking Instructions

Marking in accordance with Specification MIL–P–10025. The descriptive nomenclature shall be:

"100 Projectile Body—Metal parts assembly for shell, 81 mm Mortar, WP, M370"

The shipping classification shall be:

"Non-Explosive Ammunition"

The cubical displacement shall be 15.9 cubic feet.

5.3 **Data card.** Data card information shall be as specified in Specification MIL–A–2550.

6. NOTES

6.1 **Ballistic test.** Rounds fired for velocity-accuracy are not acceptance tests, but the results will be carefully analyzed by the Munitions Command to determine quality of ammunition produced. Evidence of excessive variation in either phase of the proving ground tests will result in immediate investigation by the Munitions Command.

6.2 **Visual examination qualification.** When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

Custodian:
Army—MU

Preparing activity:
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